Control systems: nerves and hormones

9C Threat from disease · Lesson 3 of 7

Date _____ Class _____

Today I can

- ☐ I can describe how the nervous system detects a stimulus and produces a response.
- ☐ I can describe how hormones affect the body and compare the nervous and hormonal systems.

What happens today

1. Do Now: the hot pan
2. Learn: The nervous system · Picking up a pen · Hormones: chemical messengers · and more
3. Practical: Reaction time: the ruler drop
4. Challenge, then the exit ticket

Student Book pages 42 to 43 · Workbook pages 31

What I learned today (one or two sentences)

Key Vocabulary

Term	Definition
nervous system	the organ system that senses changes and makes the body respond rapidly
central nervous system	the brain and spinal cord (CNS)
nerve	an organ that carries impulses between the CNS and the rest of the body
stimulus	a change that can be detected, e.g. light, sound, touch
receptor	a cell in a sense organ that detects a stimulus
impulse	an electrical signal that travels along a nerve
effector	a muscle or gland that carries out a response
hormone	a chemical messenger made in a gland and carried in the blood
gland	an organ that makes and releases a substance, e.g. a hormone
target organ	an organ that responds to a particular hormone
insulin	a hormone from the pancreas that makes cells take up glucose from the blood

Quick Review

the hot pan

You touch a hot pan and pull your hand away before you even feel the pain.

1 Which organ detected the heat?

2 Which organ moved your hand?

3 Which organ system connects them?

Core 1 of 3

Everyone does these. Answer on the card or under it.

1 Match each sense organ to the stimulus it detects. Draw a line.

eye	chemicals in the air
ear	light
skin	chemicals in food
tongue	sound
nose	temperature and touch

2 Complete the text using words from the word bank.

The chemical messengers in our body are called _____. They are made in _____ and released into the _____ to travel round the body. They cause _____ organs to change what they are doing.

Core 2 of 3

Everyone does these. Answer on the card or under it.

3 Ruler drop: record the distance the ruler fell before you caught it. Then calculate the mean.

a) Complete the table.

Try	Distance the ruler fell (cm)
1	
2	
3	
4	
5	
mean	

b) Which control system did you use to catch the ruler? Give a reason.

Core 3 of 3

Everyone does these. Answer on the card or under it.

4 Number the statements 1 to 6 to show the path that impulses take when you pick up a pen.

Statement	Order
the brain processes the information	
touch receptors in the skin of the fingers detect the pen	
impulses travel along a nerve to the spinal cord and brain	
the muscles contract and lift the hand	
impulses travel along a nerve to the muscles in the arm	
the pen touches the fingers (the stimulus)	

Challenge 1 of 2

When Core is finished. Full sentences.

C1 You tread on something sharp. Suggest which control system is better for responding to this stimulus. Explain your answer, naming the receptor, the parts of the nervous system and the effector involved.

Challenge 2 of 2

When Core is finished. Full sentences.

C2 Sultan Kosen grew to 251 cm tall. Chandra Dangi was 55 cm tall as an adult. Suggest which two glands may not have been working normally in these men, and explain your reasoning.

Exit Ticket

Hand this card in at the end of the lesson.

Name _____

1 Name the two organs of the central nervous system.

2 What is a hormone, and how does it travel round the body?

3 Give one difference between a nervous response and a hormonal response.

Today I can: Goal 1 ☐ Goal 2 ☐